task1.py task1_output.txt task2.py

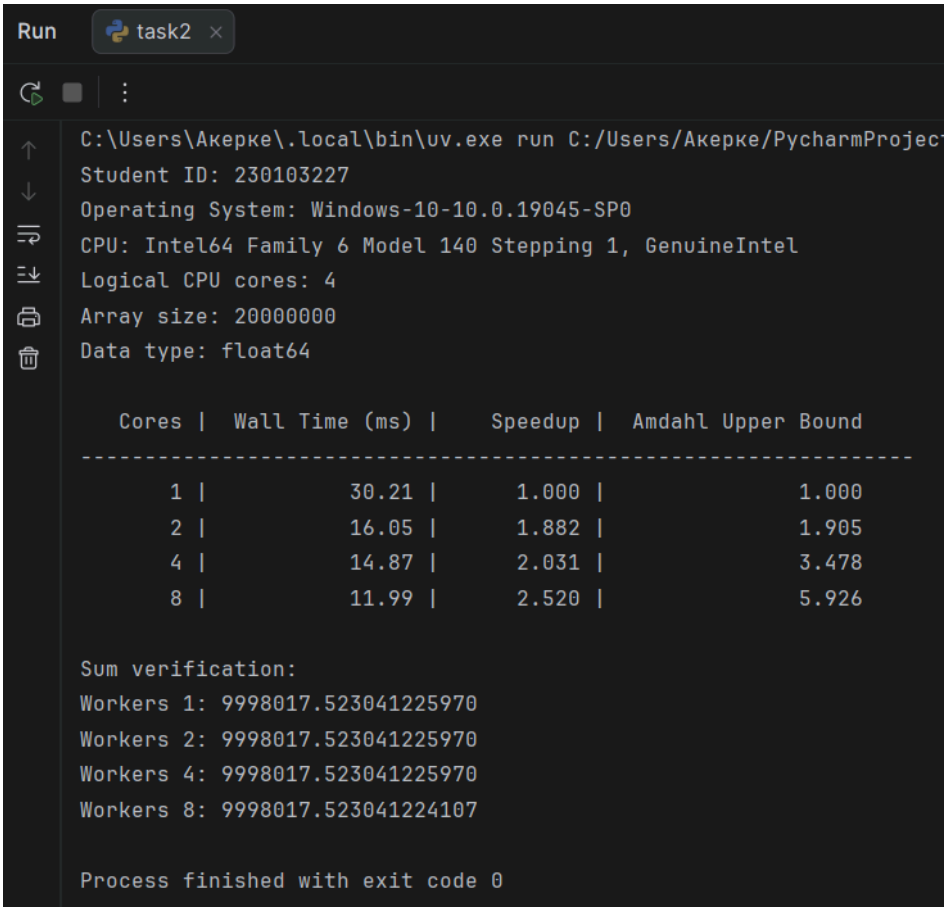
```python
import os
import platform
import time
import numpy as np
from concurrent.futures import (ThreadPoolExecutor)

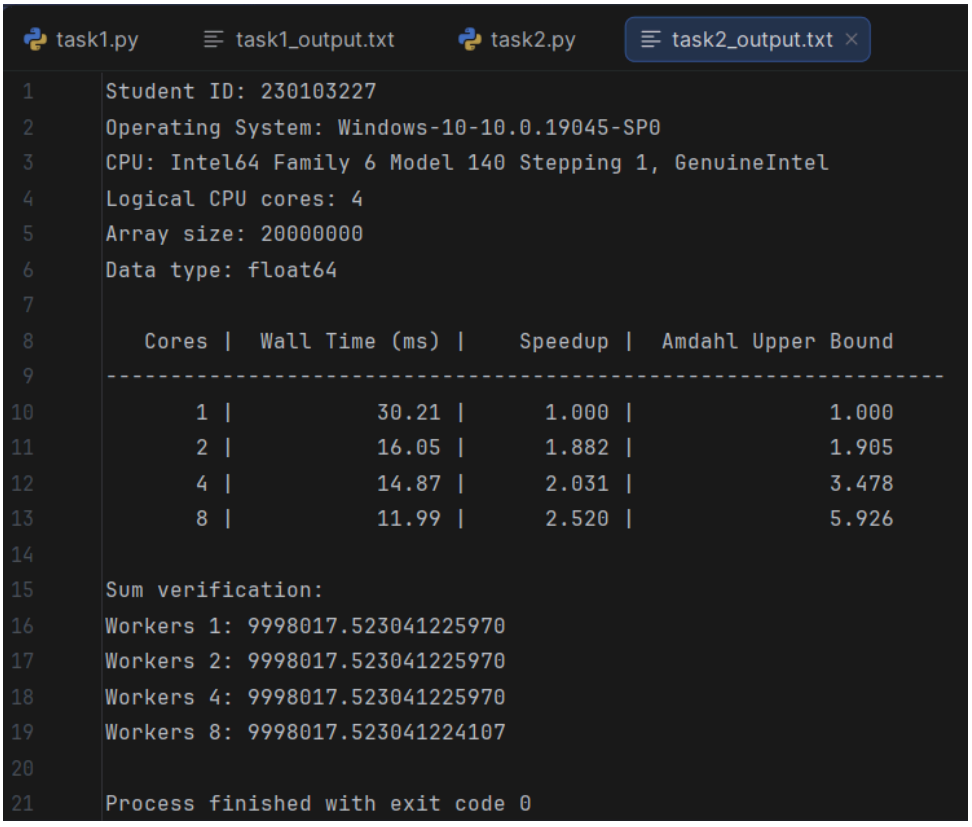
STUDENT_ID = "230103227"
ARRAY_SIZE = 20_000_000
P = 0.95
max_cores = os.cpu_count() or 1
worker_counts = sorted(set([1, 2, 4, max_cores, max_cores * 2]))
rng = np.random.default_rng(int(STUDENT_ID[-3:]))
data = rng.random(ARRAY_SIZE, dtype=np.float64)
def partial_sum(chunk):  1 usage
    return np.sum(chunk, dtype=np.float64)

def parallel_sum(workers):  1 usage
    chunks = np.array_split(data, workers)
    with ThreadPoolExecutor(max_workers=workers) as executor:
            results = list(executor.map(partial_sum, *iterables: chunks))
    return sum(results)
print("Student ID:", STUDENT_ID)
print("Operating System:", platform.platform())
print("CPU:", platform.processor() or platform.machine())
print("Logical CPU cores:", max_cores)
print("Array size:", ARRAY_SIZE)
print("Data type:", data.dtype)
print()
results = []
for workers in worker_counts:
```

```python
for workers in worker_counts:
    start = time.perf_counter()
    total_sum = parallel_sum(workers)
    end = time.perf_counter()
    elapsed = end - start

    results.append({
        "workers": workers,
        "time": elapsed,
        "sum": total_sum
    })
baseline_time = next(r["time"] for r in results if r["workers"] == 1)
print(f"{'Cores':>8} | {'Wall Time (ms)':>15} | {'Speedup':>10} | {'Amdahl Upper Bound':>19}")
print("-" * 65)
for result in results:
    workers = result["workers"]
    elapsed = result["time"]
    speedup = baseline_time / elapsed
    amdahl_bound = 1 / ((1 - P) + P / workers)
    print(
        f"{workers:8d} | "
        f"{elapsed * 1000:15.2f} | "
        f"{speedup:10.3f} | "
        f"{amdahl_bound:19.3f}"
    )
print()
print("Sum verification:")
for result in results:
    print(f"Workers {result['workers']}: {result['sum']:.12f}")
```


Run
task2
C:\Users\Акерке\.local\bin\uv.exe run C:/Users/Акерке/PycharmProjec
Student ID: 230103227
Operating System: Windows-10-10.0.19045-SP0
CPU: Intel64 Family 6 Model 140 Stepping 1, GenuineIntel
Logical CPU cores: 4
Array size: 20000000
Data type: float64
Cores | Wall Time (ms) | Speedup | Amdahl Upper Bound
1 | 30.21 | 1.000 | 1.000
2 | 16.05 | 1.882 | 1.905
4 | 14.87 | 2.031 | 3.478
8 | 11.99 | 2.520 | 5.926
Sum verification:
Workers 1: 9998017.523041225970
Workers 2: 9998017.523041225970
Workers 4: 9998017.523041225970
Workers 8: 9998017.523041224107
Process finished with exit code 0


task1.py
task1_output.txt
task2.py
task2_output.txt
Student ID: 230103227
Operating System: Windows-10-10.0.19045-SP0
CPU: Intel64 Family 6 Model 140 Stepping 1, GenuineIntel
Logical CPU cores: 4
Array size: 20000000
Data type: float64
Cores | Wall Time (ms) | Speedup | Amdahl Upper Bound
1 | 30.21 | 1.000 | 1.000
2 | 16.05 | 1.882 | 1.905
4 | 14.87 | 2.031 | 3.478
8 | 11.99 | 2.520 | 5.926
Sum verification:
Workers 1: 9998017.523041225970
Workers 2: 9998017.523041225970
Workers 4: 9998017.523041225970
Workers 8: 9998017.523041224107
Process finished with exit code 0